NAME:
HABIB ID:

LINEAR ALGEBRA

SPRING 2026

QUIZ 2

Max Marks: 10

Time: 10 minutes

Q. Show that if $\boldsymbol{B}$ has a column of zeros and $\boldsymbol{A}$ is any matrix for which $\boldsymbol{AB}$ is defined, then $\boldsymbol{AB}$ also has a column of zeros.

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QUIZ 3

Max Marks: 10

Time: 10 minutes

Q. Consider the system of equations below:

$$x = 0$$

$$-y + \frac{1}{2}z = 0$$

$$x + 2y = 2$$

(a) By setting up an appropriate augmented matrix, evaluate the inverse of the coefficient matrix of the system. [6]

(b) Is the system consistent or inconsistent? Why? [1]

(c) Hence, find a solution to the system of equations. [3]

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QUIZ 2

Max Marks: 10

Time: 10 minutes

Q. Show that if $\boldsymbol{AB}$, $\boldsymbol{BC}$ and $\boldsymbol{AC}$ are defined, then $\boldsymbol{B}$ is square matrix.

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QUIZ 3

Max Marks: 10

Time: 10 minutes

Q. Consider the system of equations below:

$$-y = 4$$
$$x = 0$$
$$2x + y + z = -1$$

(a) By setting up an appropriate augmented matrix, evaluate the inverse of the coefficient matrix of the system. [6]

(b) Is the system consistent or inconsistent? Why? [1]

(c) Hence, find a solution to the system of equations. [3]

Solution to Quiz 02

Question 1

Show that if B has a column of zeros and A is any matrix for which AB is defined, then AB also has a column of zeros.

Solution

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then the product AB is defined and is an $m \times p$ matrix.

Suppose that the j-th column of matrix B is a zero column, that is,

$$\text{column } j \text{ of } B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

By the definition of matrix multiplication, the j-th column of AB is given by

$$\text{column } j \text{ of } AB = A \times (\text{column } j \text{ of } B).$$

Since the j-th column of B is the zero vector, we have

$$A \times \mathbf{0} = \mathbf{0}.$$

Therefore, the j-th column of AB is also a zero column.

$$\boxed{\text{Hence, } AB \text{ has a column of zeros.}}$$

OR

Proof. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Since AB is defined, these dimensions are compatible.

Assume that the j-th column of B is a column of zeros. That is,

$$b_{1j} = b_{2j} = \cdots = b_{nj} = 0.$$

We claim that the j-th column of the matrix AB is also a column of zeros.

Let $(AB)_{ij} = c_{ij}$ be an arbitrary entry in the j-th column of AB. By the definition of matrix multiplication,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Since the j-th column of B is a zero column, we have

$$b_{1j} = b_{2j} = \cdots = b_{nj} = 0.$$

Therefore,

$$c_{ij} = a_{i1}(0) + a_{i2}(0) + \cdots + a_{in}(0) = \sum_{k=1}^{n} a_{ik}(0) = 0.$$

Since i was arbitrary, every entry in the j-th column of AB is zero. Hence, the j-th column of AB is a column of zeros.

$\boxed{\text{Therefore, if } B \text{ has a column of zeros, then } AB \text{ also has a column of zeros.}}$

OR

In general, if $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix.

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2p} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix}$$

the entry $(AB)_{ij}$ in row i and column j of AB is given by

$$(AB)_{ij} = c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ik}b_{kj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Let the j-th column of B is zero, so we have $b_{1j} = b_{2j} = \cdots = b_{nj} = 0$.

$$c_{ij} = a_{i1}0 + a_{i2} + \cdots + a_{jn}0 = \sum_{k=1}^{n} a_{jk}0 = 0.$$

$$\boxed{\text{Hence, the } j\text{-th column of } AB \text{ is a zeros , i.e } \sum_{j=1}^{p}\sum_{k=1}^{n} a_{ik}b_{kj}}$$

Question 2

Show that if AB, BC, and AC are defined, then B is a square matrix.

Solution

Assume that

A is an $m \times n$ matrix, $\quad B$ is a $p \times q$ matrix, $\quad C$ is a $r \times s$ matrix.

Step 1: Since AB is defined, the number of columns of A must equal the number of rows of B:

$$n = p.$$

Step 2: Since BC is defined, the number of columns of B must equal the number of rows of C:

$$q = r.$$

Step 3: Since AC is defined, the number of columns of A must equal the number of rows of C:

$$n = r.$$

Step 4: From Steps 1 and 3, we get

$$p = r.$$

Using this in Step 2,

$$q = p.$$

Hence,

$$p = q.$$

Conclusion: Since matrix B is of order $p \times q$ and $p = q$, matrix B is a square matrix.

$$\boxed{B \text{ is a square matrix.}}$$

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SOLUTION QUIZ 3A

So, augment the matrix with the identity matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & \frac{1}{2} & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

1 mark for setting up the correct augmented matrix

1 marks for each E.R.O [4 marks]

1 mark for arriving at and picking out the correct inverse

Subtract row 1 from row 3: $R_3 = R_3 - R_1$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & \frac{1}{2} & 0 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$$

Multiply row 2 by -1: $R_2 = -R_2$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$$

Subtract row 2 multiplied by 2 from row 3: $R_3 = R_3 - 2R_2$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right]$$

Add row 3 multiplied by $\frac{1}{2}$ to row 2: $R_2 = R_2 + \frac{R_3}{2}$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right]$$

We are done. On the left is the identity matrix. On the right is the inverse matrix.

(b) Since the inverse of the coefficient matrix exists, the system is consistent i.e. a solution exists [1 mark]

(c) A**x** = **b**

x = A^{-1}**b** = $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ 1 mark for correctly setting up the product of matrices

2 marks for correct multiplication + final answer

deduct marks accordingly for errors as they occur

full marks in part c if their answer to part a was incorrect but they have used their answer for all further fully correct calculations

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SOLUTION QUIZ 3B

Swap the rows 1 and 2:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Subtract row 1 multiplied by 2 from row 3: $R_3 = R_3 - 2R_1$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \end{array}\right]$$

Multiply row 2 by -1: $R_2 = -R_2$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \end{array}\right]$$

Subtract row 2 from row 3: $R_3 = R_3 - R_2$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

We are done. On the left is the identity matrix. On the right is the inverse matrix.

1 mark for setting up the correct augmented matrix

1 marks for each E.R.O [4 marks]

1 mark for arriving at and picking out the correct inverse

(b) Since the inverse of the coefficient matrix exists, the system is consistent i.e. a solution exists [1 mark]

(c) A**x = b**

x = A^{-1}**b** = $\begin{pmatrix} 0 \\ -4 \\ 3 \end{pmatrix}$ 1 mark for correctly setting up the product of matrices

2 marks for correct multiplication + final answer

deduct marks accordingly for errors as they occur

full marks in part c if their answer to part a was incorrect but they have used their answer for all further fully correct calculations